

increased number of customer reported incidents without the increased headcount.

What is wrong here?

The very first problem here is 'retrofit thinking.' Acme Solutions is starting with a solution in mind and then trying to fit in with their current issue of growing customer base and incident reports. They formed their hypothesis while already choosing the solution, which is the fundamental mistake in hypothesis framing.

Instead, if they focus on the problem first, the solution may be different. If Acme acknowledges that an increase in customer base is leading to increased incident reports, there is a fundamental flaw in their service or product. There is something inherent in their service or product that is causing incidents in the first place, and the best way to address it is to eliminate the cause of incidents.

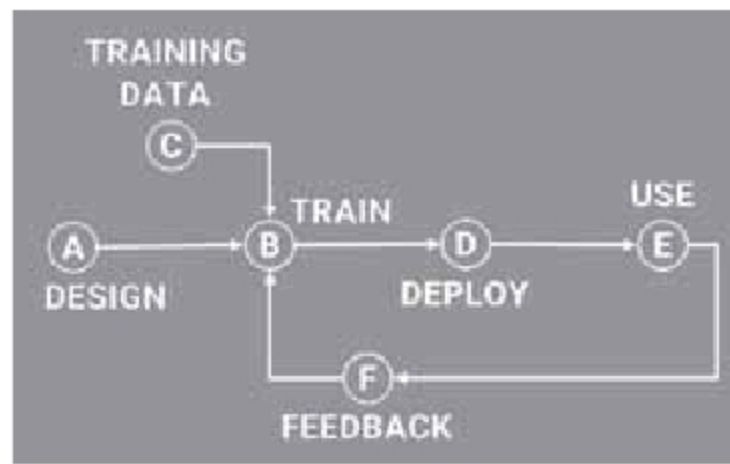
Once Acme realises that solving flaws in product/service would eliminate or reduce the number of reported incidents, their need for increased headcount is likely to go away. When that happens, the need for AI chatbot will go away too.

In this case, if Acme were to go ahead with an AI chatbot, their incident count will not reduce. However, they will be able to respond to more customers that are reporting incidents. Now, no matter how intelligent and efficient this chatbot is, once customers start to report an incident, it means that they are already unhappy and frustrated with Acme's product and service. Acme's brand loyalty has already taken a hit. Solving customer problems would not restore it; at best, it will stop it from getting worse.

You will notice that AI risks do not necessarily start with the AI system or application itself. These risks would have already crept in, well before the solution building starts or deployment begins.

Therefore correctly framing the hypothesis, validating it with the data, and supporting it with statistical analysis is highly essential. It should be your very first checkpoint in evaluating the risks of the AI solution.

Let us use the accompanying figure, which shows six broad stages of the AI solution lifecycle. Using this



The six broad stages of AI solutions' lifecycle

as a reference to ask seven key questions will ensure your AI solution's success.

1. Are you addressing the right root cause?

Once you frame the correct hypothesis and validate that you indeed need an AI system to fix something for you, it is time to decide what exactly needs fixing?

Typically, we can establish a relationship between the problem and its cause in the following manner:

$$y = f(x)$$

where y is the problem and x is the cause or driver (or multiple causes or drivers) of that problem, and f represents the process or formula that makes x result in y.

If we continue with our earlier Acme Solutions' example, it may look something like this:

Incident reports = f(poor product manual, low-quality product, etc)

It means incident reports are caused by poor product manuals, which customers cannot understand and therefore create an issue. It also means that sometimes the product manual may not be the only issue; poor product build may also cause an incident report.

So, if your solution was to address the increasing incident report problem, it cannot do anything to incident reports directly. It will have to work on the product manual, product build, and other contributing factors. When those factors are improved, incident reports will reduce automatically.

What if by implementing an AI-based transcriber, the product manual quality improves?

It would be a useful AI solution that is set to address a valid root cause. Your AI solution must be able

to address one or more of the root causes of your main problem. It ensures that your issue will be solved.

If an AI solution is not addressing any of the root causes or is addressing only low-impact root causes, there is a risk, and you must fix it.

2. Is the solution correctly trained?

One of the critical stages in AI solution development is the training of the system.

If you are working with a sophisticated solution, it usually follows two-stage learning. In case of a simple AI solution, it happens in only one stage, that is, during the model building phase.

The training stage is where the foundation is formed for the first time, and the solution is trained using those models. Therefore the accuracy of these models is highly dependent on the training dataset.

If the data fed during the training of the solution is not good enough, it can result in several problems later on, and the risk would manifest in different forms.

For example, algorithmic bias may seep into the solution, which will affect all the outcomes. Machine learning (ML) algorithms identify patterns in the data and then build a predictive model based on that pattern. Any rule created or a decision taken by the AI solution is dependent on that. If those patterns have any bias in them, the solution will amplify that bias and may produce outcomes, which in turn may reinforce the bias further.

The risk with such invisible bias is that when the AI solution makes any errors, it is tough to diagnose and pinpoint the source of an issue.

If you suspect the training quality of your AI solution, or have doubts over the data used for training, the solution has a higher risk of problems. On the other hand, if you are confident about the source data and the training process followed, the risk is quite low.

There is not one and the absolute correct way to train the AI system; variations exist. If the confidence level in those training procedures and training data is low, the solution carries high-